KPS PUBLIC SCHOOLS • STUDENT TECH BENCH PROGRAM

CTL Chromebook PX121E

Solder-Free Repair Guide



Every field-replaceable repair that needs a screwdriver, not a soldering iron.

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Built from CTL's published PX121E specs and parts catalog. CTL has not published a PX121E service manual; screw counts and connector locations must be confirmed on the bench's first teardown (see the notes boxes).

1 Read first

This guide covers every PX121E repair the Tech Bench can do with hand tools: swapping a part, reseating a cable, or fixing software. Anything that needs a soldering iron, hot air, or board-level work is listed in Section 6 and goes to CTL or the district lead tech instead.

> **How this guide was built.** CTL publishes a PX121E parts catalog (33 listings) and repair videos for its other PX models, but as of October 2026 it has not published a PX121E service manual. The procedures here follow the parts CTL sells and the way CTL's PX-series machines come apart. Screw counts and connector positions are left as fill-in boxes. Fill them in the first time you do each repair.

Bench levels

Every repair carries a level badge. Students work up through the levels as a supervisor signs off on their training.

Level	Who may do it	Typical repairs
LEVEL 1	Any trained student	Software resets, single keys, hinge caps, base cover, battery
LEVEL 2	Student with supervisor sign-off on this repair	Display, bezel, cables, IO board, camera, hinges
LEVEL 3	Lead tech or adult supervisor only	Keyboard + C cover (top case swap), liquid damage, mainboard swap

Before you touch any device

1. ☐ Check the asset tag and look the device up in the district inventory. Note any open ticket.
2. ☐ Ask the lead tech whether the unit is still under CTL warranty or an accidental-damage plan. CTL includes 5-day OEM service with its Lifecycle Services, and a warranty RMA may cost the district less than a part.
3. ☐ Photograph all four sides of the device before you open it. Damage you didn't cause should be on record.
4. ☐ Record the symptom in the student's or teacher's own words on the ticket.
5. ☐ Back up nothing: ChromeOS stores student work in the cloud. Do warn the user that local Downloads may be lost on a reset.

Safety rules

- **Power off, unplug, then disconnect the battery** before any work under the base cover. A Chromebook in sleep is still powered.
- **Lithium batteries:** never puncture, bend, or pry against the battery. A swollen, hot, hissing, or smoking battery is a Level 3 stop. Move it to the metal battery bin or sand, away from people, and get an adult.
- **ESD:** wear a grounded wrist strap on an ESD mat. Touch only the edges of boards. Static you can't feel can still kill a chip.
- **Glass and sharp edges:** a cracked LCD panel has glass edges. Wear safety glasses when removing a cracked screen.
- **Only plastic tools touch ribbon cables and clips.** Metal picks are for screws only.

- **No liquids near open devices** except 90%+ isopropyl alcohol for cleaning, used sparingly.

Tools and supplies

Item	Use	Notes
Precision screwdriver, Phillips #0 and #00	All case screws	Magnetic tip helps. Confirm size on first teardown.
Plastic spudgers and opening picks (6+)	Releasing clips, lifting connectors	Replace when the edges chip
Plastic tweezers (ESD-safe)	Handling FFC ribbon cables	Never metal tweezers on connectors
Magnetic screw mat or labeled tray	Keeping screws in order	Draw the base outline and place screws where they came from
ESD mat + wrist strap	Static protection	Clip the strap to the mat's ground point
90%+ isopropyl alcohol + soft brush	Cleaning connectors, spill residue	Not for the LCD face
Kapton tape	Re-taping cables and connectors	Replace any tape you remove
Microfiber cloth	Cleaning panels and bezels	
Recovery USB (8 GB+)	ChromeOS recovery	Made with Chromebook Recovery Utility
Known-good USB-C charger (45 W)	Testing power and charging	CTL's 45 W USB-C adapter fits
Safety glasses	Cracked-screen removal	
Fire-safe battery bin or sand bucket	Damaged batteries	Required at every bench

2 Know the machine

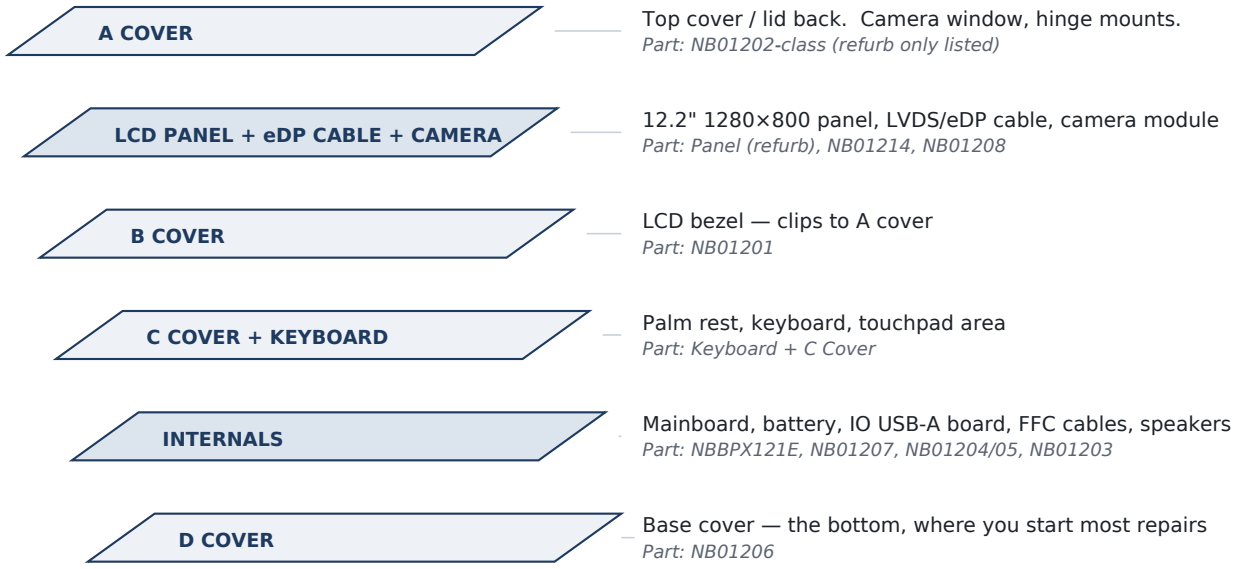
Specs that matter for repair

Item	PX121E	Why it matters
Display	12.2" LCD, 16:10, 1280×800, non-touch	Only the 12.2" PX121E panel fits. PX11E/PX111E panels are 11.6" and will not.
Processor	Intel N150 (N-series)	Soldered to mainboard — no repair
Memory / storage	8 GB LPDDR5 / 64 GB eMMC	Both soldered. Storage failure = mainboard.
Battery	41 Wh internal	Replaceable part, NBBPX121E
Ports	2× USB-C 3.1, 1× USB-A 3.1, microSD, lock slot, audio	USB-A is on the replaceable IO board
Wireless	Wi-Fi 6E (Intel AX211), Bluetooth 5.3	Not sold as a separate PX121E part
Camera	720p, front-facing, privacy shutter	Replaceable module, NB01208
Keyboard	IP41 spill-resistant, single-key repair	Individual keycaps pop off and on
Hinge	180° lay-flat	Left and right sold separately
Durability	MIL-STD-810H, 75 cm drop tested	
Weight	About 2.65 lb (1.2 kg)	
Updates	ChromeOS auto-updates to June 2035	Worth repairing for years
Model / SKU	PX121E, CTL SKU CBUS1200001	Check the label on the D cover

Specs from CTL's PX121E product page and launch materials. Notebookcheck's launch coverage listed a 45 W battery; CTL's own page says 41 Wh, which this guide uses.

Anatomy: the A/B/C/D covers

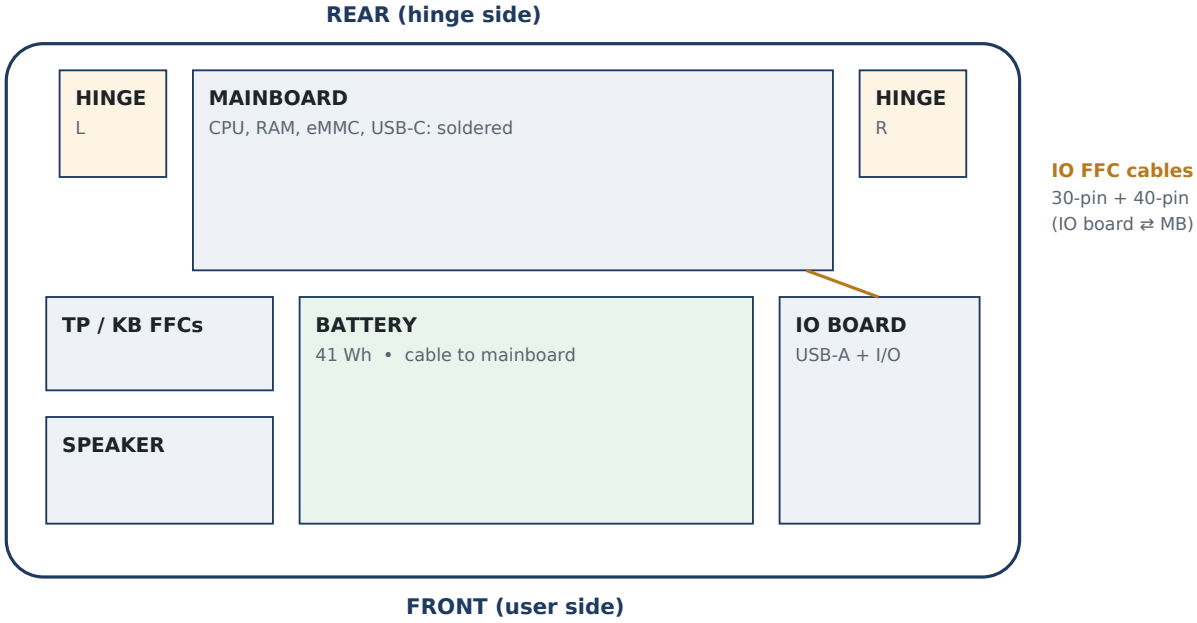
CTL, like most laptop makers, names the shell pieces by letter. Parts are ordered by these names, so learn them.



Top of stack = lid exterior. Bottom of stack = underside of laptop. Labels follow CTL's A/B/C/D naming.

Inside the base

Once the D cover is off you see the parts below. Use this as a rough map, then correct it on your first teardown.



ILLUSTRATIVE ONLY — block positions are typical of CTL PX-series bases, not measured from a PX121E.

PX121E parts list

These are the PX121E parts CTL lists on ctl.net as of October 2026. Most come new or refurbished. Stock changes often. Districts with a CTL account manager can order through them.

Part (CTL listing name)	SKU	Repair
PX121E Battery	NBBPX121E	R4
PX121E Keyboard + C Cover	—	R15
PX121E 12.2" LCD Panel (refurbished listed)	—	R10
PX121E LVDS Cable (display cable)	NB01214	R11
PX121E LCD Bezel (B Cover)	NB01201	R9
PX121E Top Cover (A Cover) (refurbished listed)	—	R14
PX121E Base Cover (D Cover)	NB01206	R3
PX121E Left Hinge / Right Hinge	NB01210 / NB01211	R13
PX121E Left Hinge Cap / Right Hinge Cap	NB01212 / NB01213	R2
PX121E Camera Module	NB01208	R12
PX121E IO USB/A Board (9C-N1KZC0010)	NB01207	R6
PX121E IO FFC 30P / 40P cables (IO board to MB)	NB01204 / NB01205	R7
PX121E CP FFC 8P cable (touchpad)	NB01203	R8
PX121E touchpad (not listed: ask CTL)	—	R18
PX121E speakers (not listed: ask CTL)	—	R19
CTL 45 W USB-C Adapter	—	Triage
Rugged Flex Case for PX121E	—	Prevention

Not in the catalog. CTL does not list a PX121E mainboard, speakers, touchpad, Wi-Fi card, or individual PX121E keycaps for sale as of October 2026. CTL sells keycaps for the PX11E/PX141E series; do not assume they fit the PX121E. Ask your CTL account manager or support (800-642-3087, support@ctl.net) for parts that aren't listed.

Stocking suggestion. Hinge caps, the base cover, and the FFC cables are small, cheap, and the most-used parts. Keep a few on hand so a small repair never waits on shipping. Batteries and Keyboard + C Covers are worth stocking once you see your failure rate.

3 Triage: symptom → repair

Always start with the cheapest, least invasive fix. Work down the "try first" column before you order parts.

Symptom	Try first	Then	Part likely needed
Won't turn on, no lights	Known-good charger 30 min → S1 hard reset	R5 reseat battery cable → R4 battery	Battery, or mainboard (Sec. 6)
Not charging / charge light off	Other USB-C port, other charger → S1	S2 battery health → R4	Battery; USB-C port = mainboard
Battery drains fast or shows low health	S2 Diagnostics battery test	R4	Battery
Swollen battery / base bulging	STOP — Level 3	R4 by lead tech only	Battery
Black screen, power light on	External display via USB-C; S1	R5 reseat eDP cable → R11 → R10	Cable or panel
Lines, flicker, changes with lid angle	S1	R11 display cable	Display cable
Cracked or blotchy screen	—	R10 LCD panel	LCD panel
Bezel loose or broken	Re-seat clips	R9	B cover
Lid cracked / back of lid damaged	—	R14	A cover
Hinge stiff, loose, or lid wobbles	Check hinge screws	R13	Hinge(s)
Hinge cap missing or cracked	—	R2	Hinge cap
One key missing or broken	—	R1	Keycap
Several keys dead, or a row dead	S2 keyboard test; S1	R5 reseat KB cable → R15	Keyboard + C cover
Touchpad dead or erratic	S1; check touchpad not disabled in settings	R8 touchpad cable → R18 touchpad	TP cable or touchpad
USB-A port or microSD dead	Try another device in port; S1	R7 IO cables → R6 IO board	IO board or cables
USB-C port dead	Other port, other charger	Sec. 6	Mainboard (soldered port)
Camera not detected / black	Check privacy shutter; S1	R12	Camera module
No sound	Headphones work? Volume, output; S1; S3	R19 speakers	Speakers (ask CTL)
Wi-Fi or Bluetooth missing	S1 → S3 recovery	Sec. 6	Ask CTL
Liquid spill	Power off NOW, unplug	R16 (Level 3)	Depends on damage

Symptom	Try first	Then	Part likely needed
Slow, crashing, error screens	S1 → S3 powerwash/recovery	Sec. 6 if recovery fails	Often none
"ChromeOS is missing or damaged"	S3 recovery	Sec. 6 if recovery fails	eMMC = mainboard
Base cover cracked, feet missing	—	R3	D cover

4 Software repairs

These need no tools and fix a surprising share of tickets. Always do S1 before opening a device.

S1 Hard reset (EC reset) LEVEL 1

Fixes	Won't power on, won't charge, frozen, black screen, unresponsive keyboard or touchpad
Time	2 minutes
Tools	Charger
Data loss	None

A hard reset restarts the embedded controller, the small chip that manages power, the keyboard, and charging. It does not erase anything.

Procedure

1. ☐ Unplug the charger and any USB devices.
2. ☐ Hold the power button for 10 seconds to make sure the device is fully off.
3. ☐ Press and hold **Refresh** (the circular-arrow key, usually F3 position), then tap **Power**.
4. ☐ Release Refresh when the device starts.
5. ☐ If it does not start, plug in a known-good 45 W USB-C charger, wait 30 minutes, and repeat steps 2–4 with the charger connected.

Verify before closing the ticket

- Device powers on and reaches the login screen.
- Charging light comes on with the charger connected.

> **Tip.** After a battery has been disconnected (R4, R5), many Chromebooks won't wake on the power button until a charger is plugged in. Plug in before you decide it's dead.

S2 Built-in hardware diagnostics LEVEL 1

Fixes	Confirms the fault before you order a part: battery health, CPU, memory, keyboard, touchpad, network
Time	5–15 minutes
Tools	None
Data loss	None

Procedure

1. ☐ Sign in with a bench test account (not the student's account if you can avoid it).
2. ☐ Open the **Launcher**, search for **Diagnostics**, and open it.
3. ☐ **System** tab: read battery health and cycle count. Run the battery discharge and charge tests with the charger as instructed.
4. ☐ Run the CPU and memory tests.
5. ☐ **Connectivity** tab: check that Wi-Fi is detected and run the tests.

6. ☐ **Input** tab: open the keyboard tester and press every key. Test the touchpad.
7. ☐ Take a screenshot (Ctrl + Show windows) of any failed result and attach it to the ticket.

Verify before closing the ticket

- Results recorded on the ticket.
- Part ordered only for a test that failed or a fault you saw yourself.

Diagnostics app layout varies a little between ChromeOS versions. If a tab has moved, look for the same test by name.

S3 Powerwash and USB recovery — LEVEL 1

Fixes	Slow, crashing, corrupted OS, "ChromeOS is missing or damaged"
Time	Powerwash 10 min; recovery 20–40 min
Tools	Recovery USB (8 GB+) for recovery only
Data loss	**Yes.** Erases everything stored locally (Downloads, local files).

District-managed devices. KPS Chromebooks are enrolled in the district's Google Admin console. A powerwash or recovery on an enrolled device normally forces it to re-enroll at the first Wi-Fi connection. Confirm with the lead tech that forced re-enrollment is on before you wipe, and that the bench Wi-Fi can reach enrollment. Never enter Developer Mode on a district device.

Procedure

1. ☐ **Powerwash first.** At the login screen press Ctrl + Alt + Shift + R, select Restart, then Powerwash. (On managed devices this may be blocked by policy. If so, use recovery.)
2. ☐ **Recovery (if powerwash fails or is blocked):** on a working computer, install the Chromebook Recovery Utility Chrome extension and create a recovery USB for the PX121E. Search the utility's model list for PX121E or pick it from a list.
3. ☐ On the PX121E, hold **Esc + Refresh**, then tap **Power**. Release when the recovery screen appears.
4. ☐ Insert the recovery USB when asked and follow the prompts.
5. ☐ When it reboots, connect to Wi-Fi and confirm the device re-enrolls to the KPS domain.

Verify before closing the ticket

- Device boots, reaches the enrollment or KPS sign-in screen.
- Device shows as enrolled in the Admin console.
- If recovery reports an error writing to storage, the eMMC has failed: escalate (Sec. 6).

If the Recovery Utility lists no PX121E image, CTL can provide one. Use CTL's "Chromebook SHIM Request" support form or contact CTL support.

5 Hardware repairs

Most hardware repairs start with R3 (base cover off) and R5 step 1 (battery disconnected). Those steps are written out once and referenced everywhere else. Work in this order and you rarely break anything.

Golden rules for every hardware repair. Photograph each layer before you remove it. Keep screws in a mat laid out like the device. Release every connector latch before pulling a cable. Never force a part: if it doesn't move, find the screw or clip you missed. Reassemble in reverse order and test before the last screws go in.

R1 Single keycap replacement — LEVEL 1

Fixes	Missing, cracked, or loose key. Key that doesn't click or sticks.
Part	Replacement keycap (ask CTL for the PX121E keycap — PX11E/PX141E caps are not confirmed to fit)
Time	5 minutes
Tools	Plastic pick, tweezers

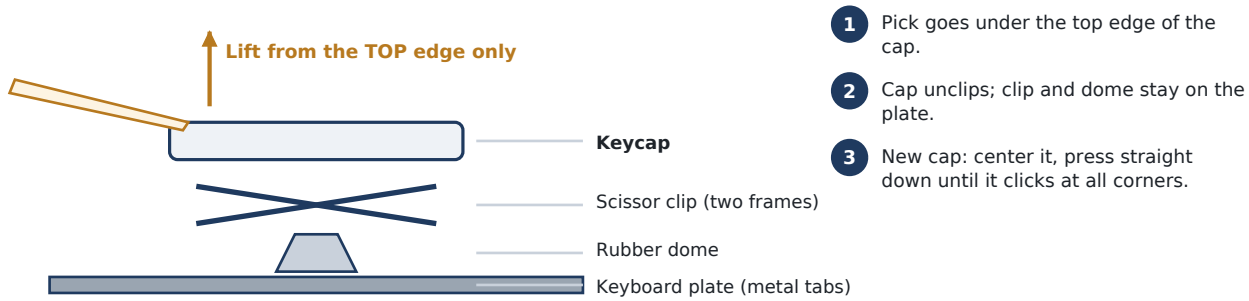


Illustration: side view of one key. Not to scale.

CTL designed the PX121E keyboard for single-key repair. Each key has a cap, a scissor clip (two interlocking plastic frames), and a rubber dome.

Procedure

1. ☐ Power off the device. Leave the lid fully open on a flat surface.
2. ☐ Slide a plastic pick under the top edge of the damaged cap and lift gently until the cap unclips. Then lift it free.
3. ☐ Look at the scissor clip. If it's broken or missing, it must be replaced too: ask CTL whether caps ship with clips.
4. ☐ To fit a new clip: hook the clip's pins under the small metal tabs on the keyboard plate, then press it flat.
5. ☐ Line up the new cap over the clip. Press straight down on the center until you hear it click on all corners.
6. ☐ For wide keys (Space, Shift, Enter, Backspace): hook the metal stabilizer wire into its holders before pressing the cap down.

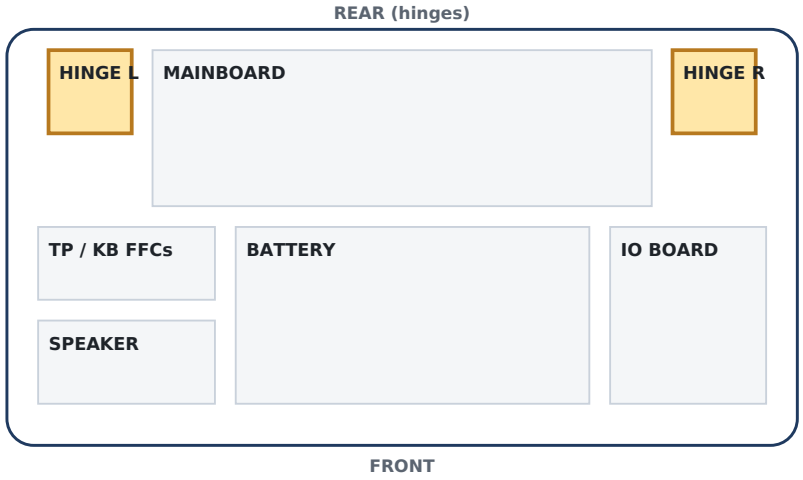
Verify before closing the ticket

- Key clicks evenly and springs back.
- Key registers in the Diagnostics keyboard tester (S2).
- Neighboring keys still work.

Spare caps. When a keyboard is scrapped for other reasons, harvest its caps and clips. Bag and label them as PX121E.

R2 Hinge cap replacement — LEVEL 1

Fixes	Missing, cracked, or loose plastic cover over a hinge
Part	PX121E Left Hinge Cap NB01212 or Right Hinge Cap NB01213
Time	5–10 minutes
Tools	Plastic spudger; Phillips driver if screwed



1. Caps cover the hinges at the rear corners.
2. Look for a screw first (outside, or under the D cover).
3. Pry at the seam with a plastic spudger.
4. Check the hinge underneath before fitting the new cap.

Illustration: base interior, hinge areas highlighted. Positions illustrative.

Procedure

1. ☐ Power off. Open the lid to about 90°.
2. ☐ Look under and behind the cap for a screw. If you find one, remove it.
3. ☐ Pry gently at the seam with a plastic spudger and work the cap off its clips. If it won't release, check under the D cover for a hidden screw (R3).
4. ☐ Clean the area. Check the hinge underneath: if it's cracked or loose, do R13 now.
5. ☐ Line up the new cap and press it on until all clips seat. Refit any screw.

Verify before closing the ticket

- Cap sits flush with no gap.
- Lid opens and closes through the full 180° without rubbing.

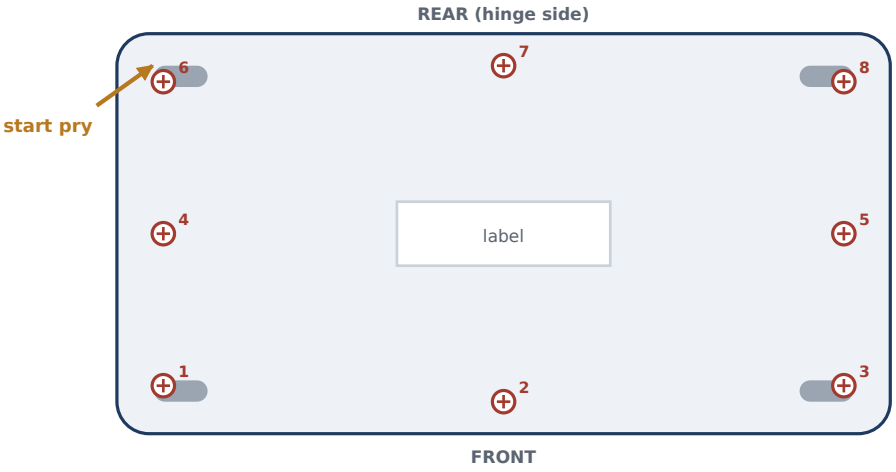
First-teardown notes — fill in the first time you do this repair

Cap held by: clips only / screw(s)? Where? _____

R3 Base cover (D cover) removal and replacement

LEVEL 1

Fixes	Cracked or missing base cover, missing feet. Also the first step for R4–R17.
Part	PX121E Base Cover (D Cover) NB01206 — only if replacing
Time	10 minutes
Tools	Phillips driver, plastic picks, screw mat



1. Remove all base screws. Count them.
2. Lay screws on the mat in this same pattern.
3. Pry from the corner shown, then work around the edge.
4. Disconnect the battery before anything else.

Illustration: underside. Screw count and positions not yet confirmed for the PX121E.

Procedure

1. ☐ Power off completely (hold Power 10 s). Unplug the charger and everything else.
2. ☐ Close the lid and flip the device over onto a soft, clean mat with the hinge facing away from you.
3. ☐ Remove all base screws. Place each on the screw mat in the same position. Screws can be different lengths, so this matters.
4. ☐ Look for hidden screws under rubber feet or labels if the cover won't release. Do not peel the asset label.
5. ☐ Insert a plastic pick at the seam near a hinge corner. Slide it along the edge to release the clips, one side at a time.
6. ☐ Lift the D cover from the hinge side toward the front. Go slowly: the cover may still be caught on a clip at the front.
7. ☐ **Stop here and do R5 step 1 (disconnect the battery) before anything else.**
8. ☐ To refit: line up the front edge, lower the cover, press around the edge until every clip snaps, then refit screws in their original holes. Snug only. Do not overtighten into plastic.

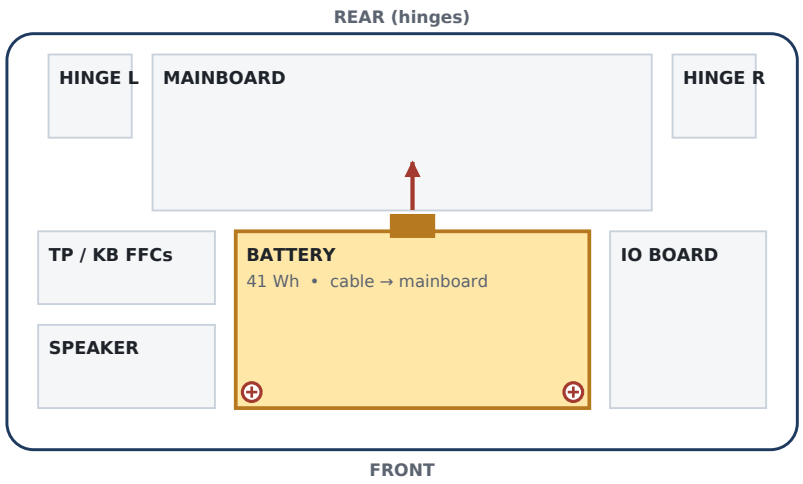
Verify before closing the ticket

- No gaps around the seam. Device sits flat without rocking.
- All screws refitted, none left over.

First-teardown notes — fill in the first time you do this repair	
Number of base screws	_____
Screw lengths and where each goes	_____
Hidden screws under feet? (Y/N, where)	_____
Best starting point for pry	_____

R4 Battery replacement — LEVEL 2

Fixes	Won't hold charge, low health in S2, swollen battery, won't power on without charger
Part	PX121E Battery NBBPX121E
Time	15–20 minutes
Tools	Phillips driver, plastic spudger, ESD strap



1. Unplug the battery cable from the mainboard first (red arrow).
2. Lift-up or slide-out plug: check which before pulling.
3. Remove the battery screws (count them).
4. Lift out flat. Never bend or pry against the cell.

Illustration: base interior, battery highlighted. Screw positions illustrative.

Swollen battery = Level 3. If the base is bulging or the battery is puffy, do not press on it or pry it. The lead tech handles it and the old cell goes straight to the fire-safe bin.

Procedure

1. ☐ Do R3 to remove the base cover.
2. ☐ Find the battery cable where it plugs into the mainboard. Lift the connector straight up with a plastic spudger, rocking it gently side to side. Some connectors slide horizontally out of their socket; check which kind you have and note it in the box below.
3. ☐ Remove the battery screws. Note how many and their length.
4. ☐ Lift the battery out. If adhesive holds it, slide a plastic card under it slowly. **Never bend the battery.**
5. ☐ Check that the new battery's part label matches. Place it in the same position.
6. ☐ Refit the battery screws, then plug the cable into the mainboard. Press down evenly until it seats fully.

7. ☐ Refit the D cover (R3 step 8).
8. ☐ Plug in the charger. The device may need the charger before it will power on (see S1 tip).

Verify before closing the ticket

- Charging light comes on.
- Device powers on and shows a battery percentage.
- Run the S2 battery test. Health should read near 100% for a new battery.
- Old battery is in the recycling bin, not the trash. CTL offers free battery recycling through Call2Recycle.

First-teardown notes — fill in the first time you do this repair	
Battery connector type: lift-up / slide-out	_____
Number of battery screws and length	_____
Adhesive under battery? (Y/N)	_____

R5 Reseating internal connectors — LEVEL 2

Fixes	Dead keyboard or touchpad, black screen, dead USB-A, no power after a drop. Drops often unseat connectors.
Part	None (unless a cable is damaged)
Time	15–25 minutes
Tools	Plastic spudger, plastic tweezers, isopropyl alcohol

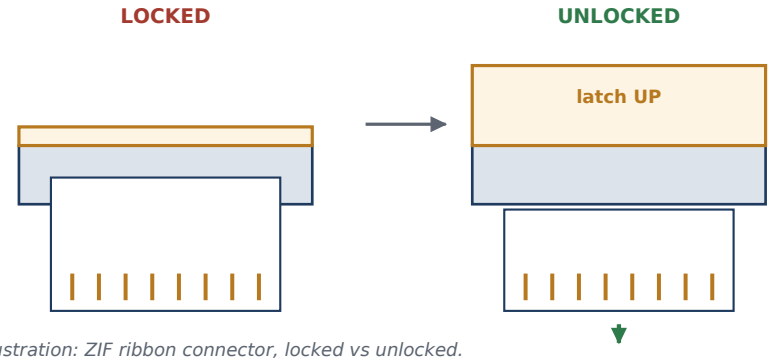


Illustration: ZIF ribbon connector, locked vs unlocked.

1. Flip the hinged latch up (or slide a drawer-type latch out) with a spudger.
2. Ribbon now slides out with zero force. Never pull a locked ribbon.
3. Refit: contacts facing the same way, push in straight and fully, close latch.
4. Board-to-wire plugs (battery, speaker): pry at the plug body, never the wires.

Two kinds of connectors live inside. Learn both before you start:

Type	How it releases	How it reconnects
Board-to-wire (battery, eDP display cable, speakers)	Lift straight up, or slide out horizontally. Pry at the connector body, never the wires.	Line up and press flat until fully seated.
ZIF / FFC (flat ribbon cables: IO board, touchpad, keyboard)	Flip up the small hinged latch (or slide it out). Then the ribbon slides out with no force.	Slide ribbon fully in, contacts facing the correct way, then close the latch.

Procedure

1. ☐ **Disconnect the battery first** (R4 step 2). This is step 1 for every repair under the base cover.
2. ☐ Hold the power button for 5 seconds with the battery disconnected to drain leftover power.
3. ☐ Work through each connector related to the symptom. Photograph it, release it, and inspect it for bent pins, burn marks, or a creased ribbon.
4. ☐ If a contact looks dirty or corroded, clean it with a brush dampened in isopropyl alcohol. Let it dry fully (5 minutes).
5. ☐ Reconnect each one. For ZIF ribbons, check the cable sits straight and all the way in before closing the latch. A crooked ribbon is the #1 cause of "it still doesn't work."
6. ☐ Reconnect the battery last. Plug in the charger and test **before** refitting the D cover.

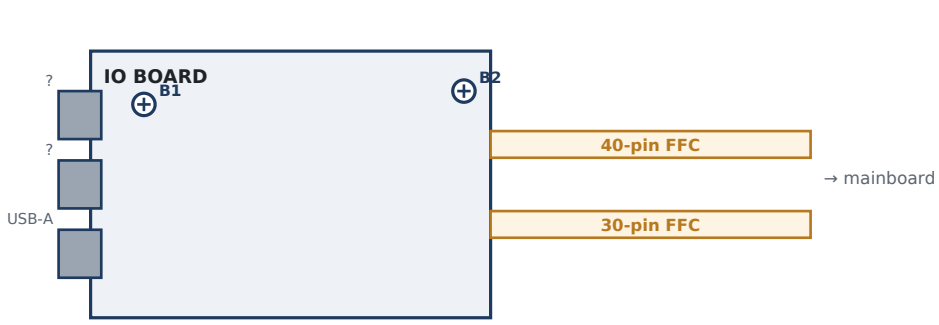
Verify before closing the ticket

- The symptom is gone in a full test, not just at boot.
- If a ribbon is creased or torn, replace it (R7, R8, R11).

R6 IO USB-A board replacement

LEVEL 2

Fixes	Dead USB-A port, and other ports on the IO board (often microSD or audio — confirm which on first teardown)
Part	PX121E IO USB/A Board NB01207 (9C-N1KZC0010)
Time	20 minutes
Tools	Phillips driver, plastic spudger, tweezers



1. Unlock both ribbon connectors (30P and 40P), both ends.
2. Slide out both ribbons.
3. Remove the IO board screws (count them).
4. Tilt the board inward to clear the port openings.

Illustration: USB board from above. Port order and screw spots illustrative. Confirm on yours.

The IO board is a small daughterboard joined to the mainboard by two flat ribbons (30-pin and 40-pin). Replacing it fixes the ports it carries with no soldering. Try R7 first: a ribbon is far cheaper than the board.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ Find the IO board at the side of the base with the USB-A port.
3. ☐ Release the latches on both FFC ribbons (30P and 40P) and slide them out, at the IO board end and the mainboard end.

4. ☐ Remove the IO board screws.
5. ☐ Lift the board out. It may be held by the port lip; tilt it inward to clear the case opening.
6. ☐ Fit the new board with its ports lined up in the case cutouts. Refit screws.
7. ☐ Reconnect both ribbons (new ribbons if the old ones were creased) and close the latches.
8. ☐ Reconnect the battery, test, then refit the D cover.

Verify before closing the ticket

- USB-A reads a flash drive and charges a phone.
- Every other port on the IO board works (microSD card, headphones if present).

First-teardown notes — fill in the first time you do this repair

Which ports are on the IO board?	_____
Number of IO board screws	_____
Which ribbon is 30P, which is 40P?	_____

R7 IO board FFC ribbon cable replacement

LEVEL 2

Fixes	IO board ports dead or intermittent, especially after a drop. Creased or torn ribbon.
Part	IO FFC 30P cable NB01204 and/or IO FFC 40P cable NB01205
Time	15 minutes
Tools	Plastic spudger, plastic tweezers

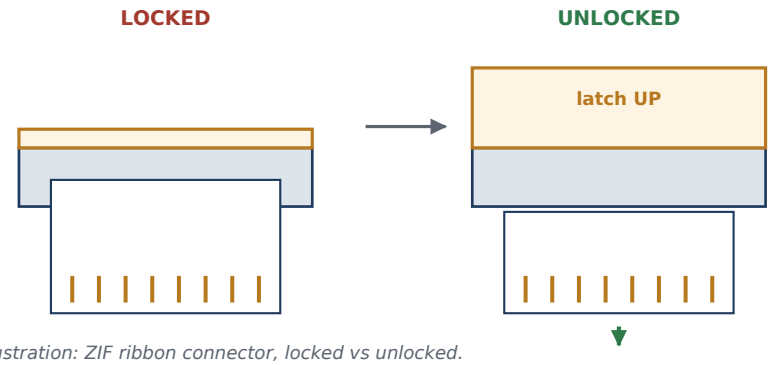


Illustration: ZIF ribbon connector, locked vs unlocked.

1. Flip the hinged latch up (or slide a drawer-type latch out) with a spudger.
2. Ribbon now slides out with zero force. Never pull a locked ribbon.
3. Refit: contacts facing the same way, push in straight and fully, close latch.
4. Board-to-wire plugs (battery, speaker): pry at the plug body, never the wires.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ Photograph the ribbon's route, including any tape holding it down.
3. ☐ Flip up both ZIF latches (IO board end and mainboard end) and slide the ribbon out.
4. ☐ Compare old and new: same pin count (30 or 40) and length (108 mm per CTL's listing).
5. ☐ Slide the new ribbon in, contacts facing the same way as the old one. Close both latches.
6. ☐ Replace any tape you removed.
7. ☐ Reconnect the battery and test before closing up.

Verify before closing the ticket

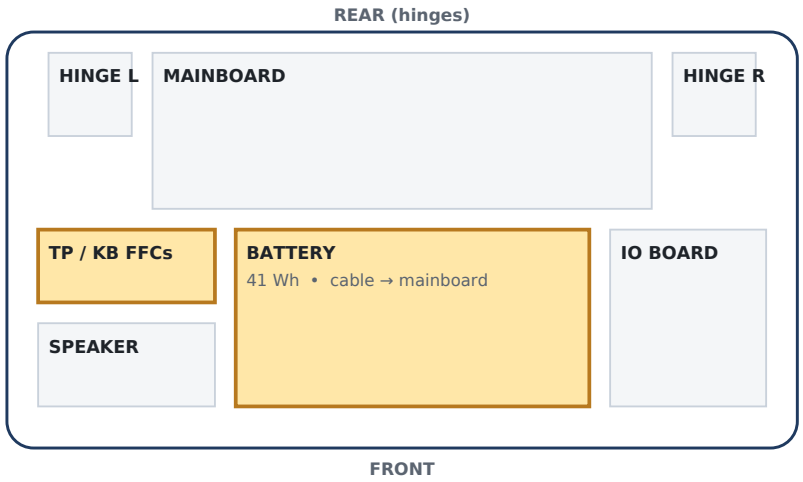
- All IO board ports work.
- Ribbon lies flat with no kink.

Cheap insurance: when replacing either ribbon, replace both. It saves a second repair.

R8 Touchpad cable replacement

LEVEL 2

Fixes	Touchpad completely dead or erratic, with keyboard working
Part	PX121E CP FFC 8P touchpad cable NB01203
Time	15–20 minutes
Tools	Plastic spudger, plastic tweezers



1. 8-pin touchpad ribbon runs to the mainboard.
2. It may pass under the battery; remove it if so.
3. Unlock both latches before sliding the ribbon out.
4. New ribbon: contacts same way, latches closed.

Illustration: base interior. Ribbon area and route illustrative.

Check first: Settings > Device > Touchpad is enabled, and S1 doesn't fix it.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ Find the 8-pin ribbon running from the touchpad (under the palm rest) to the mainboard. It may run under the battery. If so, remove the battery (R4 steps 3–4).
3. ☐ Release both ZIF latches and remove the ribbon.
4. ☐ Fit the new ribbon, contacts the same way. Close the latches.
5. ☐ Refit the battery if removed, reconnect it, and test.

Verify before closing the ticket

- Touchpad moves the cursor, clicks, and two-finger scrolls.
- Tested in the S2 Input tab.

If a new cable doesn't fix it, the touchpad itself has failed. CTL does not list a separate PX121E touchpad, so ask CTL, or replace the Keyboard + C Cover (R15) if it includes one.

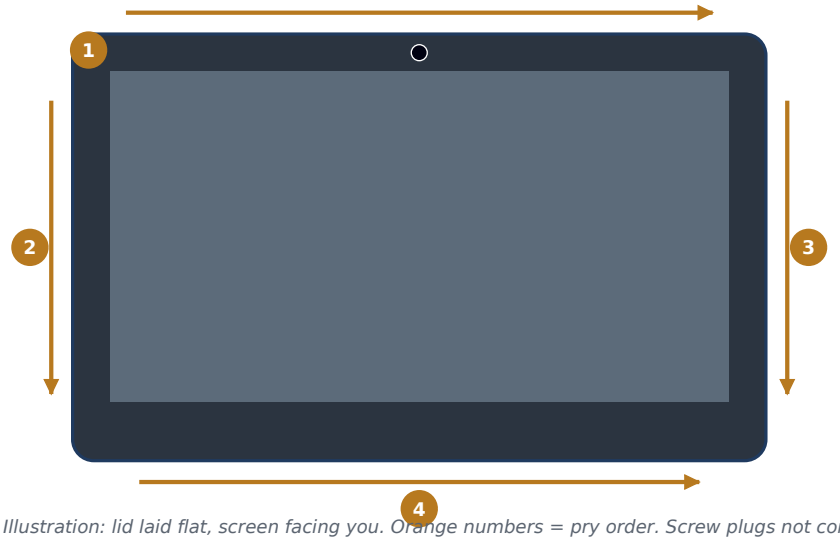
First-teardown notes — fill in the first time you do this repair

Does the touchpad cable run under the battery? (Y/N) _____

R9 LCD bezel (B cover) replacement

LEVEL 2

Fixes	Cracked, loose, or missing screen bezel. Also the first step for R10–R12.
Part	PX121E LCD Bezel (B Cover) NB01201
Time	10 minutes
Tools	Plastic picks



1. Check corners and bottom edge for rubber plugs hiding screws.
2. Start prying at a corner with a plastic pick.
3. Work along each side, releasing clips one at a time.
4. Do the bottom (hinge) edge last.
5. Don't pull on the camera window area.

Illustration: lid laid flat, screen facing you. Orange numbers = pry order. Screw plugs not confirmed on PX121E.

Clips, not glue (usually). CTL's PX-series bezels typically clip on. If you meet strong resistance or adhesive, stop and check: warming the edge gently with a hair dryer on low softens tape. Never use a heat gun.

Procedure

1. ☐ Power off and unplug. Open the lid to 180° flat on the mat, screen facing up.
2. ☐ Check the bezel corners and bottom edge for rubber plugs covering screws. Remove any plugs and screws.
3. ☐ Insert a plastic pick between the bezel and the A cover at the bottom center. Twist gently to release the first clip.
4. ☐ Work your way around the edge with a second pick, releasing clips one at a time. Do the bottom (hinge) edge last.
5. ☐ Lift the bezel off. Do not pull on the camera window area: the camera sits right behind it.
6. ☐ To fit: line up the top edge with the camera window, press around the edge until every clip snaps.

Verify before closing the ticket

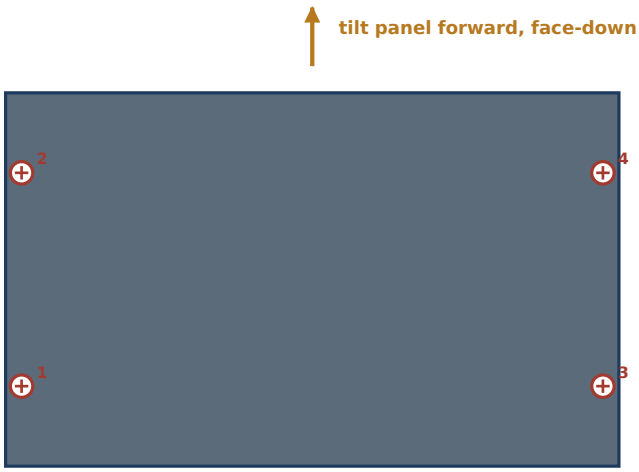
- Bezel flush all around with no gap.

- Camera window lined up; privacy shutter slides freely.
- Lid closes fully.

First-teardown notes — fill in the first time you do this repair	
Bezel held by: clips / adhesive / screws?	_____
Any screws? Where?	_____

R10 LCD panel replacement — LEVEL 2

Fixes	Cracked screen, dead pixels, ink blots, black screen that a cable swap didn't fix
Part	Refurbished PX121E 12.2" LCD Panel. **Must be the PX121E 12.2" panel.**
Time	25–30 minutes
Tools	Phillips driver, plastic picks, safety glasses, Kapton tape



1. Remove the 4 panel screws (sides of the panel).
2. Lay the panel face-down on the keyboard (cloth under it).
3. On the back: peel the tape over the connector, open the clip, pull the cable straight out.
4. Connect the new panel and TEST before screwing it down.

Illustration: lid with bezel removed. Screw positions are typical. Count them on yours.

Cracked glass. Wear safety glasses. Put tape across a shattered panel before handling it so shards stay together.

Test first: plug a USB-C monitor in. If the external display works, the mainboard is fine and the fault is the panel or cable.

Procedure

1. ☐ Power off and unplug. Do R3 and disconnect the battery. (Prevents a power surge into the new panel.)
2. ☐ Do R9 to remove the bezel.
3. ☐ Remove the panel screws (often two to four along the sides or bottom). Some panels are held by adhesive strips instead. Pull the tabs slowly if so.
4. ☐ Tilt the panel forward, face down onto the keyboard, onto a microfiber cloth.
5. ☐ On the back of the panel, peel back the tape over the display cable connector. Flip up the latch (if any) and pull the connector straight out of the panel.
6. ☐ Connect the cable to the new panel. Press it fully in, close the latch, and re-tape it with Kapton.

7. ☐ Reconnect the battery and power on **before** screwing the panel in. Check the picture.
8. ☐ Power off, disconnect the battery, then refit the panel screws, bezel (R9), and D cover (R3).

Verify before closing the ticket

- Full image with no lines, dead pixels, or uneven backlight.
- Brightness keys work across the full range.
- Image stays steady while you move the lid through its full range.

First-teardown notes — fill in the first time you do this repair	
Panel held by: screws / adhesive?	_____
Number of panel screws	_____
Display connector: latch type?	_____

R11 Display (LVDS/eDP) cable replacement

LEVEL 2

Fixes	Flicker, lines, or black screen that changes with lid angle. Image fine on an external display.
Part	PX121E LVDS Cable NB01214
Time	40–50 minutes
Tools	Phillips driver, plastic picks, tweezers, Kapton tape

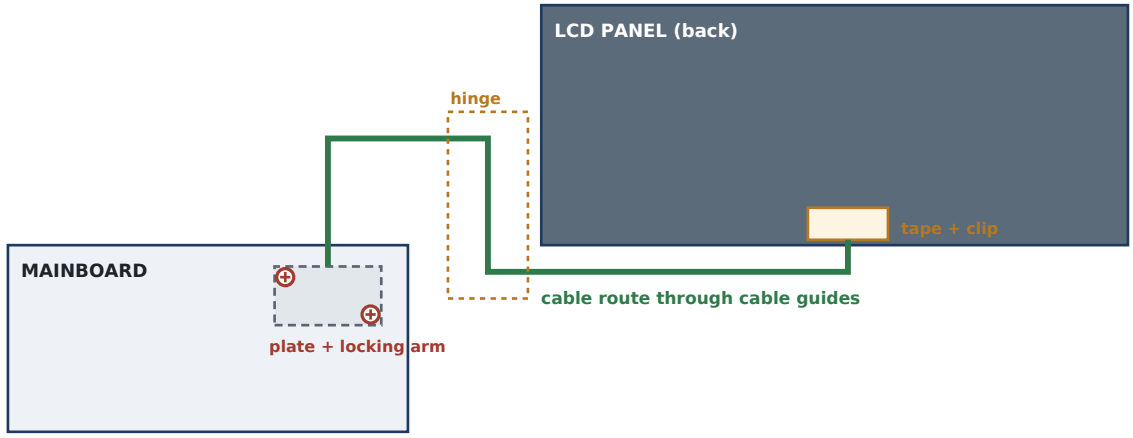


Illustration: display cable path, mainboard to panel (route simplified). Photograph the real route before removing.

CTL lists this as an "LVDS" cable. On modern panels it's technically an eDP cable. Either name means the same part: the cable from the mainboard, through the hinge, to the back of the LCD.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ At the mainboard end: peel the tape, release the display cable connector, and lift it off the board.
3. ☐ Unroute the cable from its channels and tape in the base. Photograph the route first.
4. ☐ Do R9 (bezel) and R10 steps 3–5 (panel face down, disconnected).

5. ☐ Free the cable where it runs through the hinge area. You may need to loosen the hinge or remove the hinge cap (R2). If the camera shares this cable, disconnect it at the camera module (R12).
6. ☐ Route the new cable exactly along the old path. Pinched or twisted routing will fail again within weeks.
7. ☐ Connect both ends, re-tape, reconnect the battery, and test before reassembly.
8. ☐ Reassemble in reverse order.

Verify before closing the ticket

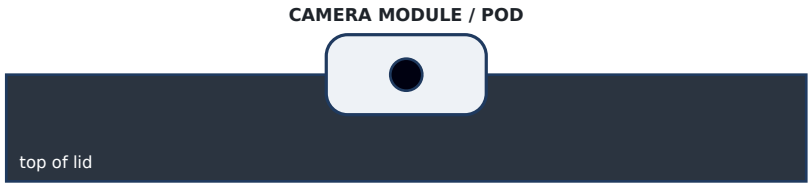
- Steady image at every lid angle from closed to 180° flat.
- Camera works (if it shares this cable).
- Open and close the lid 20 times while watching the screen.

First-teardown notes — fill in the first time you do this repair	
Does the camera connect through this cable? (Y/N)	_____
Does the cable route through a hinge? Which side?	_____

R12 Camera module replacement

LEVEL 2

Fixes	Camera not detected, black image, or broken
Part	PX121E Camera Module NB01208
Time	15–20 minutes
Tools	Plastic picks, tweezers



1. Bezel off first (it covers the camera's front).
2. Unplug the small camera cable with tweezers.
3. Lift the module by its board edges, never the lens.
4. Test in the Camera app before closing up.

Illustration: camera at the top-center of the lid. Cap shapes simplified.

Check first: is the privacy shutter open? Does the Camera app see the camera after S1? Does it work after a powerwash (S3)?

Procedure

1. ☐ Power off, unplug, and do R3 and disconnect the battery.
2. ☐ Do R9 to remove the bezel.
3. ☐ Find the camera module at the top center of the lid.
4. ☐ Disconnect its cable. Use tweezers to lift the connector straight off.

5. ☐ Lift the module out. It's often held by light adhesive. Use a plastic pick under the board, not the lens.
6. ☐ Fit the new module in the same position, lined up with the bezel's camera window. Connect the cable.
7. ☐ Reconnect the battery and test the Camera app before refitting the bezel.

Verify before closing the ticket

- Camera app shows a clear, focused image.
- Privacy shutter fully blocks the lens when closed.
- Camera works in a Google Meet test call.

R13 Hinge replacement (left or right) LEVEL 2

Fixes	Loose or floppy lid, lid that won't stay up, cracked hinge, grinding
Part	PX121E Left Hinge NB01210 or Right Hinge NB01211
Time	30–45 minutes
Tools	Phillips driver, plastic picks

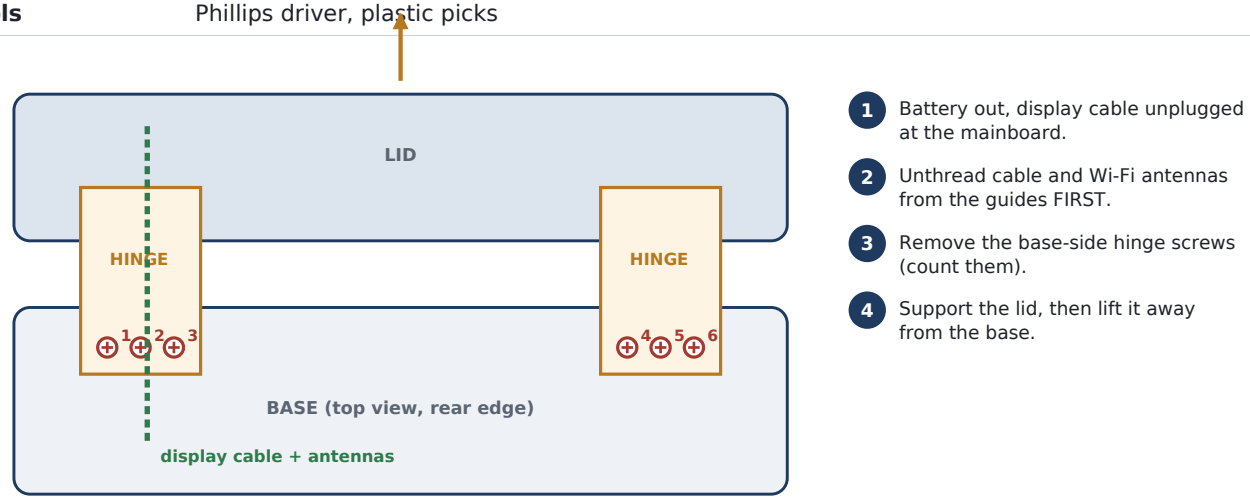


Illustration: hinge area seen from above with the D cover off. Positions approximate.

Check first: a loose lid is often just loose hinge screws. Tighten them before replacing anything. If the screw threads spin without tightening, the threaded post in the plastic is stripped. Escalate: that's a cover replacement, not a hinge.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ Do R2 to remove the hinge cap on the side you're working on.
3. ☐ Do R9 to remove the bezel. Keep the panel in place unless the hinge screws are behind it.
4. ☐ Remove the hinge screws on the lid (A cover) side. If they're behind the panel, do R10 steps 3–4.
5. ☐ Remove the hinge screws on the base side.
6. ☐ Free the display cable from the hinge if it passes through (see R11). Do not stretch it.

7. ☐ Remove the old hinge. Compare it to the new one: left and right are mirror images and are not interchangeable.
8. ☐ Fit the new hinge, base screws first, then lid screws. Snug them evenly.
9. ☐ Reconnect the battery and test the display before reassembling the bezel and cap.

Verify before closing the ticket

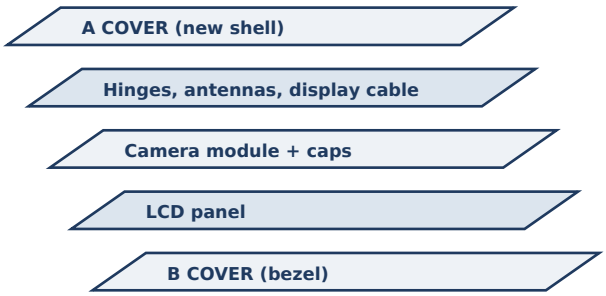
- Lid holds its position at any angle and opens to a full 180°.
- Both hinges feel equally stiff.
- Display steady through full lid travel (cable not pinched).

First-teardown notes — fill in the first time you do this repair	
Hinge screws on base side (count)	_____
Hinge screws on lid side (count)	_____
Are lid-side screws behind the panel? (Y/N)	_____

R14 Top cover (A cover) replacement

LEVEL 3

Fixes	Cracked, dented, or broken back of the lid. Stripped hinge screw posts in the lid.
Part	PX121E Top Cover (A Cover) — refurbished listed; new not listed
Time	45–60 minutes
Tools	Phillips driver, plastic picks, tweezers, Kapton tape



1. Strip the old lid top to bottom: bezel, panel, camera, hinges, cable, antennas.
2. Build the new lid in the reverse order.
3. Test the display and Wi-Fi before the bezel goes on.

Illustration: lid layers, outside (top) to inside (bottom).

This repair moves every lid part into a new shell. It combines R9, R10, R11, R12, and R13. Do it only after you've done each of those on its own.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ Do R9 (bezel off), R10 steps 3–5 (panel out and disconnected), and R12 steps 4–5 (camera out).
3. ☐ Free the display cable from the lid and hinges (R11 step 5).
4. ☐ Remove both hinges from the lid (R13 step 4). Leave them attached to the base if you can.
5. ☐ Lift the old A cover away.

6. ☐ Move any small parts that stay with the lid: antenna cables, magnets, rubber bumpers, tape. Photograph their positions.
7. ☐ Fit the new A cover onto the hinges and refit the lid-side hinge screws.
8. ☐ Refit camera, display cable route, and panel. Test before refitting the bezel.
9. ☐ Refit bezel, D cover.

Verify before closing the ticket

- Display, camera, and Wi-Fi all work (antennas often run through the lid).
- Lid closes flush and opens to 180°.
- Lid-close sleep works (the magnet is in place).

First-teardown notes — fill in the first time you do this repair	
Wi-Fi antennas routed through the lid? (Y/N)	_____
Lid magnet location	_____
Other small parts transferred	_____

R15 Keyboard + C cover replacement — LEVEL 3

Fixes	Multiple dead keys, a dead row, liquid-damaged keyboard, cracked palm rest
Part	PX121E Keyboard + C Cover
Time	60–90 minutes
Tools	Phillips driver, plastic picks, tweezers, ESD strap, screw mat, Kapton tape

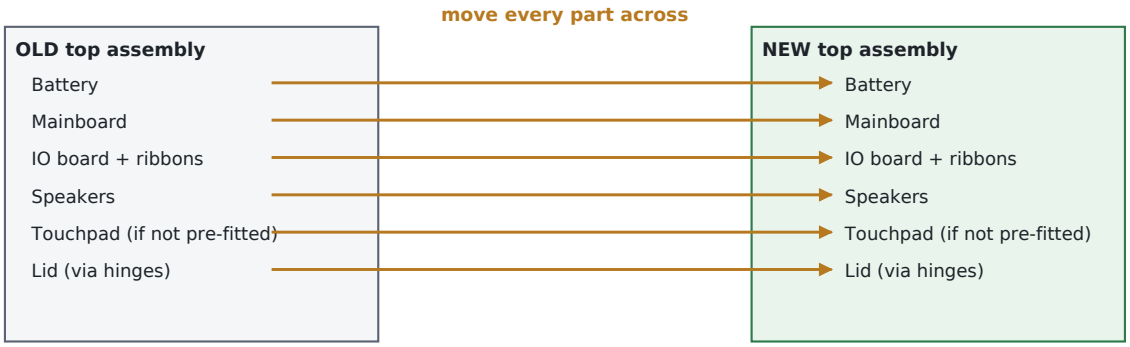


Illustration: what moves to the new top assembly. Check what comes pre-fitted first.

The keyboard comes built into the C cover, so this is a **full transplant**: every internal part moves from the old C cover to the new one. It is the largest repair in this guide. Plan for one class period or more.

Unbox and compare first. Before you start, check what's already fitted to the new C cover: touchpad? speakers? power button? Write it in the record box. Anything not included must move from the old cover.

Procedure

1. ☐ Do R3 and disconnect the battery.

2. ☐ Disconnect every connector on the mainboard: display cable, IO ribbons, touchpad ribbon, keyboard ribbon, speakers. Photograph each first.
3. ☐ Remove the battery (R4 steps 3–4).
4. ☐ Remove the IO board (R6 steps 3–5).
5. ☐ Disconnect and remove the display assembly: unscrew both hinges at the base (R13 step 5) and lift the whole lid away.
6. ☐ Remove the mainboard screws and lift the mainboard out by its edges. Set it on the ESD mat.
7. ☐ Remove speakers and any other part not already on the new C cover (touchpad, brackets, rubber feet, tape).
8. ☐ Transfer everything to the new C cover in reverse order: speakers, touchpad if needed, mainboard, IO board, battery.
9. ☐ Refit the lid at the hinges. Reconnect every cable, using your photos to check each one.
10. ☐ Reconnect the battery and test fully (keyboard, touchpad, display, ports, sound) before refitting the D cover.

Verify before closing the ticket

- Every key passes the S2 keyboard test.
- Touchpad, display, camera, ports, speakers, and Wi-Fi all work.
- No screws left over. Count against your record.

First-teardown notes — fill in the first time you do this repair	
Parts pre-fitted on new C cover	_____
Number of mainboard screws	_____
Keyboard ribbon: ZIF latch type?	_____
Total time for the first one	_____

R16 Liquid spill response

LEVEL 3

Fixes	Water, juice, soda, or other liquid on or in the device
Part	Depends on damage: often none, sometimes Keyboard + C Cover
Time	First response 10 min; drying 24–48 hours; cleaning 30–60 min
Tools	Isopropyl alcohol 90%+, soft brush, lint-free cloths

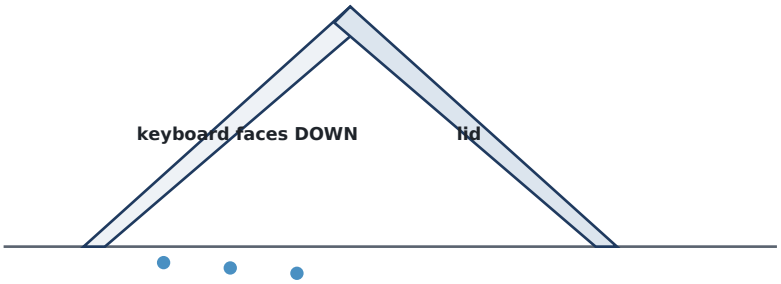


Illustration: drain position after a spill.

1. Power off (hold 10 s), unplug.
2. Tent upside down so liquid drains OUT.
3. Open and disconnect the battery ASAP.
4. Dry 24–48 h. No hair dryer, no rice.

The PX121E keyboard is IP41 spill-resistant. That means it resists light drips, not a full cup. Speed matters more than anything.

Do not power on to "check if it works." Current flowing through liquid causes the corrosion that kills boards. A device that is turned off and dried may survive. One that is switched on wet often does not.

Procedure

1. ☐ **Immediately:** hold Power 10 s to force off. Unplug the charger.
2. ☐ Turn the device upside down in a tent shape (lid open about 90°, keyboard facing down) so liquid drains out, not in.
3. ☐ Do R3 and disconnect the battery as soon as possible.
4. ☐ Inspect for liquid on the mainboard. Blot (don't wipe) with lint-free cloth.
5. ☐ For sugary or sticky liquids: clean affected areas with a soft brush and isopropyl alcohol. Sugar left behind corrodes contacts later.
6. ☐ Leave the device open, battery disconnected, to dry for at least 24 hours (48 hours if it was soaked). Do not use a hair dryer or rice.
7. ☐ Inspect for white or green corrosion on connectors and the board. Clean connectors with alcohol (R5 step 4).
8. ☐ Reconnect the battery, plug in, and run S2 fully.

Verify before closing the ticket

- Every test in S2 passes.
- No sticky keys. If keys are sticky or dead, do R15.
- Corrosion on the mainboard itself: escalate (Sec. 6).

R17 Mainboard swap (lead tech only)

LEVEL 3

Fixes	Dead USB-C ports, no power with a good battery, failed eMMC, no Wi-Fi, failed recovery
Part	PX121E mainboard — not sold on ctl.net. Obtain through CTL support or a warranty RMA, or harvest from a donor unit.
Time	60–90 minutes
Tools	Same as R15

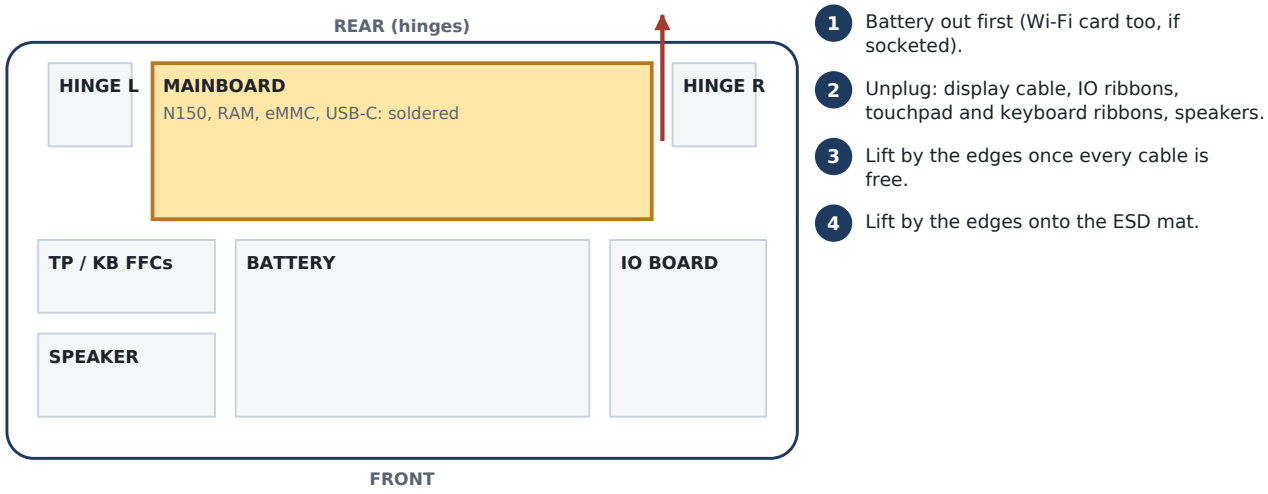


Illustration: base interior, mainboard highlighted. Red arrow = lift side.

A mainboard swap needs no soldering, but it changes the device's identity. Treat it as an administrative job as much as a repair.

Identity and enrollment. Storage, the serial number record, and the device's enrollment are tied to the mainboard. Coordinate with the lead tech before the swap: update the asset record, deprovision the old board in the Google Admin console if needed, and plan for re-enrollment. Ask CTL whether the replacement board ships with the original serial number set.

Procedure

1. ☐ Confirm the fault with S1, S2, S3, and R5 first. Most "dead mainboards" are a battery, a cable, or a connector.
2. ☐ Do R15 steps 1–6 to free the mainboard.
3. ☐ Transfer any parts mounted on the board that aren't soldered (shields, thermal pads, tape) to the new board.
4. ☐ Fit the new board and reassemble as in R15 steps 8–10.
5. ☐ Run S3 recovery if the board arrives blank, and enroll the device.

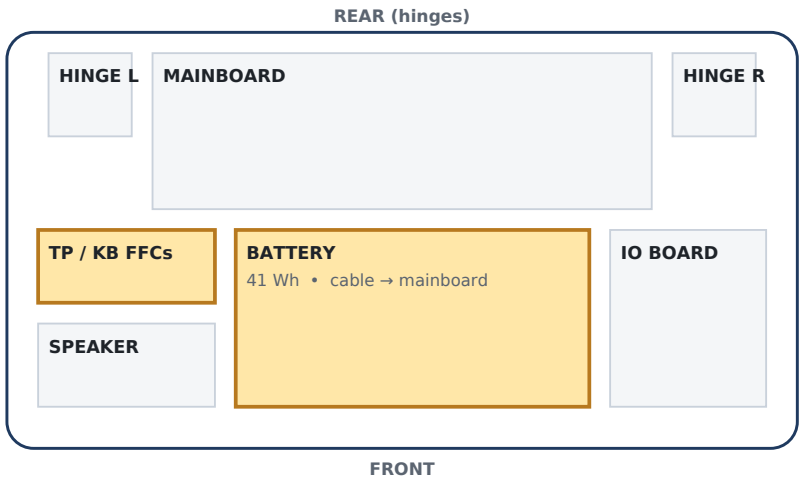
Verify before closing the ticket

- Device boots, enrolls, and passes all of S2.
- Asset record and Admin console show the correct serial number.

R18 Touchpad (trackpad) replacement

LEVEL 2

Fixes	Touchpad dead, erratic, cracked, or won't click, after a new touchpad cable (R8) didn't fix it
Part	PX121E touchpad: not listed on ctl.net. Order through CTL support or your account manager, or harvest from a donor unit.
Time	25–35 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, Kapton tape



1. Battery out first.
2. Unlock and unplug the 8-pin ribbon at the touchpad end.
3. Remove bracket screws; peel off the grounding strip.
4. Lift the pad out from the inside, never push from the top.

Illustration: base interior. Touchpad sits under the battery area; position illustrative.

Check first: touchpad turned on in Settings, S1 done, R5 reseat tried, and R8 (the much cheaper cable) replaced. Most dead touchpads are the cable.

Confirm the part first. CTL doesn't list a separate PX121E touchpad. If CTL can only supply it as part of the Keyboard + C Cover, do R15 instead.

Procedure

1. ☐ Do R3 and disconnect the battery (R4 step 2).
2. ☐ Remove the battery (R4 steps 3–4). On most PX-series bases the touchpad sits under or just in front of it.
3. ☐ Find the touchpad's 8-pin ribbon (the CP FFC from R8). Unlock the latch at the touchpad end and slide the ribbon out.
4. ☐ Peel back any mylar or tape over the touchpad bracket. Keep it intact so it can go back on.
5. ☐ Remove the touchpad bracket screws. Count them and note any lengths that differ.
6. ☐ If a grounding strip (thin foil) is attached to the touchpad, peel it off the touchpad and leave it on the frame.
7. ☐ Lift the touchpad out from the inside. Don't push on it from the palm-rest side, or the glass surface can crack.
8. ☐ Fit the new touchpad, centered in the palm-rest opening with an even gap all round. Refit the bracket screws snugly.
9. ☐ Restick the grounding strip and mylar. Reconnect the ribbon and close the latch.

10. ☐ Refit the battery, reconnect it, and test before closing the D cover.

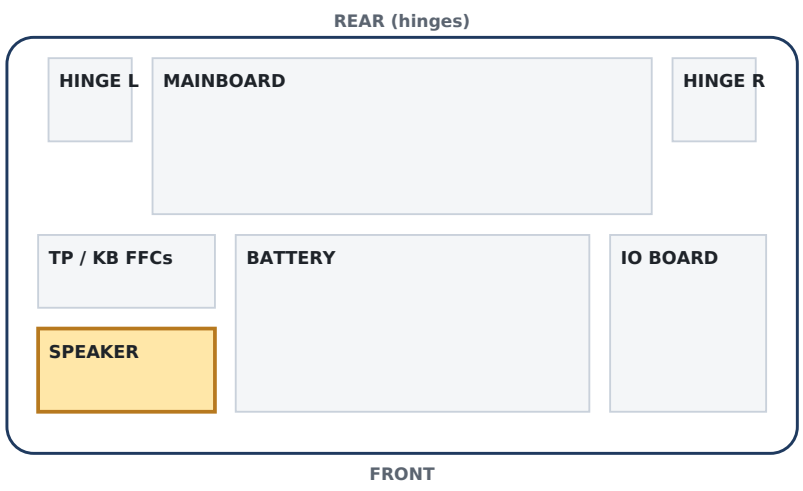
Verify before closing the ticket

- Cursor moves smoothly across the whole surface.
- Click works at all four corners, and the gap around the pad is even.
- Two-finger scroll works. Passes the S2 touchpad test.

First-teardown notes — fill in the first time you do this repair	
Touchpad bracket screws (count)	_____
Grounding strip present? (Y/N)	_____
Under the battery? (Y/N)	_____

R19 Speaker replacement — LEVEL 2

Fixes	No sound or crackling from the built-in speakers, with headphones working
Part	PX121E speakers: not listed on ctl.net. Order through CTL support or your account manager, or harvest from a donor unit.
Time	20 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers



1. Battery disconnected.
2. Unplug the speaker cable at the mainboard.
3. Unroute the cable, noting each clip.
4. Lift the speakers; keep the rubber gaskets.

Illustration: base interior, speaker highlighted. Position illustrative.

Check first: S1 and S3, then plug in headphones. If headphones also have no sound, the problem is software or the mainboard, not the speakers.

Procedure

1. ☐ Do R3 and disconnect the battery.

2. ☐ Find the speakers near the front corners of the base. Trace their cable to the mainboard.

3. ☐ Photograph the cable route, including every clip and piece of tape.

4. ☐ Unplug the speaker cable at the mainboard. Pry at the plug body, never the wires.

5. ☐ Unroute the cable from its clips. Remove any speaker screws, or lift the speakers out of their rubber mounts.
6. ☐ Check whether left and right are one part joined by the cable. They usually are, and come as a set.
7. ☐ Fit the new speakers with their rubber gaskets in place. Loose gaskets cause rattle.
8. ☐ Route the cable through every clip along the old path, then plug it into the mainboard.
9. ☐ Reconnect the battery and test before closing the D cover.

Verify before closing the ticket

- Clear sound from both sides with no rattle at full volume.
- Headphone jack still works.
- Cable isn't pinched by the D cover.

First-teardown notes — fill in the first time you do this repair	
Speaker screws or rubber mounts?	_____
Left and right one part? (Y/N)	_____

6 Out of scope: escalate

These faults need soldering, board-level repair, or parts CTL doesn't sell for this model. Students don't attempt them. Tag the device with the ticket, then send it to the lead tech for a CTL RMA or a mainboard swap (R17).

Fault	Why it's out of scope	Route
USB-C port loose, broken, or dead	USB-C ports on PX-series machines are typically soldered to the mainboard	CTL RMA or R17
Not charging with good charger, battery, and both ports tested	Charging circuit on mainboard	CTL RMA or R17
Storage (eMMC) failure — recovery can't write	eMMC is soldered	CTL RMA or R17
RAM errors in S2	LPDDR5 is soldered	CTL RMA or R17
Wi-Fi/Bluetooth hardware missing after recovery	Wi-Fi module not sold as a PX121E part	Ask CTL
Corrosion on mainboard after spill	Needs board-level cleaning or rework	CTL RMA or R17
Burn marks, melted plastic, burnt smell	Fire hazard	Lead tech immediately
Stripped screw posts in C or D cover	Requires cover replacement or threaded inserts	R15 / R3 cover swap, or CTL
Any repair that would require a soldering iron or hot air	Program policy	CTL RMA

CTL contacts

- **CTL support:** 800-642-3087 • support@ctl.net • support.ctl.net (submit a request for RMAs, parts, and the PX121E recovery image)
- **Parts store:** ctl.net/collections/chromebook-parts (filter by PX121E)
- **Repair videos and manuals for other CTL models:** support.ctl.net, "Chromebook Repair Videos and Manuals." Check back for PX121E additions. The PX111E videos are the closest relatives.

7 Sources

Facts in this guide come from these pages, checked October 2026. Procedures are the guide author's, based on CTL's parts list and standard laptop service practice. They are not CTL-issued instructions.

- CTL, "CTL Chromebook PX121E" product page — specs, 41 Wh battery, IP41 keyboard, single-key repair, AX211 Wi-Fi, ports, SKU. ctl.net/products/ctl-chromebook-px121e
- CTL, "Meet the CTL Chromebook PX121E" — MIL-STD-810H, 75 cm drop testing, Lifecycle Services. ctl.net/pages/ctl-chromebook-px121e
- CTL press release, "CTL Introduces New 12.2" Chromebook for Students," June 24, 2025 — N150 processor, updates to 2035. ctl.net/blogs/press-releases
- CTL, "CTL Genuine Chromebook Parts" — PX121E part names and SKUs. ctl.net/collections/chromebook-parts
- CTL, "PX121E Battery" product page — SKU NBBPX121E. ctl.net/products/px121e-battery
- CTL Support, "Chromebook Repair Videos and Manuals" (updated Nov 20, 2025) — repair videos for related PX models; no PX121E manual listed. support.ctl.net
- Notebookcheck, "CTL 12.2-inch Chromebook PX121E debuts..." — display resolution, ports, weight. notebookcheck.net
- ENERGY STAR certified computers listing, CTL PX121E — model certification data. energystar.gov